

Project : Calculative Foundation

Theory Concept:

1. Vectors, Operations, Distances, Norms

- A vector is a quantity that has both magnitude and direction. A vector is usually written as a list of numbers.

Example:

- If $A = (2, 3)$, it means the vector has two values.

Vector operations:

- Addition: add the corresponding values
- Subtraction: subtract the corresponding values
- Scalar multiplication: multiply every value by the same number

If $A = (2, 3)$ and $B = (1, 4)$:

$$A + B = (3, 7)$$

$$A - B = (1, -1)$$

$$2A = (4, 6)$$

Distance tells us how far two vectors are from each other.

Distance between A and B:

- $d(A,B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Simple example:

- Let $A = (1, 2)$ and $B = (4, 6)$

$$\text{Distance} = \sqrt{(4 - 1)^2 + (6 - 2)^2}$$

$$= \sqrt{3^2 + 4^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25} = 5$$

A norm means the length or size of a vector.

- $\|A\| = \sqrt{x^2 + y^2}$

Simple example:

- If $A = (3, 4)$, then

$$||A|| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$

2. Unit Vector, Dot Product, Cross Product

- A unit vector is a vector with length 1. It shows only direction.

$$u = A / ||A||$$

Simple example:

- If $A = (3, 4)$, then $||A|| = 5$

So unit vector = $(3/5, 4/5)$

- A dot product gives a single number. It helps us measure similarity between two vectors.

- $A \cdot B = a_1b_1 + a_2b_2 + a_3b_3$

Simple example:

- If $A = (1, 2)$ and $B = (3, 4)$, then

$$A \cdot B = (1 \times 3) + (2 \times 4) = 3 + 8 = 11$$

- A cross product gives a new vector. It is mainly used for 3D vectors.
- $A \times B = (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1)$

Simple example:

- If $A = (1, 0, 0)$ and $B = (0, 1, 0)$, then

$$A \times B = (0, 0, 1)$$

Dot product vs Cross product:

- Dot product gives a number
- Cross product gives a vector
- Dot product is used for angle and similarity
- Cross product is used for direction in 3D space

3. Angle Between Vectors, Projection

- The angle between two vectors tells us how much they point in the same direction.
- $\cos(\theta) = (A \cdot B) / (||A|| ||B||)$

Simple example:

- Let $A = (1, 0)$ and $B = (0, 1)$

$$A \cdot B = 0$$

$$\text{So } \cos(\theta) = 0$$

$$\text{Therefore } \theta = 90^\circ$$

This means the vectors are perpendicular.

- A projection shows how much one vector lies in the direction of another vector.
- $\text{proj}_B(A) = (A \cdot B / \|B\|^2) B$

Simple example:

- Let $A = (2, 2)$ and $B = (1, 0)$

$$A \cdot B = 2$$

$$\|B\|^2 = 1$$

$$\text{Projection of } A \text{ on } B = 2(1, 0) = (2, 0)$$

This means only the horizontal part of A is taken.

4. Matrix Types and Operations

- A matrix is a rectangular arrangement of numbers in rows and columns.

Example:

$$\begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 4 \end{bmatrix}$$

- Common matrix types:
- Row matrix: only one row
- Column matrix: only one column
- Square matrix: same number of rows and columns

- Identity matrix: diagonal values are 1, others are 0
- Zero matrix: all values are 0
- Diagonal matrix: only diagonal values may be non-zero

Matrix operations:

- Addition
- Subtraction
- Multiplication
- Transpose

For addition:

- $A + B = [a_{ij} + b_{ij}]$

For multiplication:

- $C_{ij} = \sum (a_{ik} b_{kj})$

Simple example of addition:

- If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Then

$$A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

Simple example of multiplication:

- If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} 4 & 4 \\ 10 & 8 \end{bmatrix}$$

5. Eigenvalue and Eigenvector

- An eigenvector is a special vector that does not change its direction when a matrix is applied to it. Only its length changes.
- The number that changes its size is called the eigenvalue.

$$A x = \lambda x$$

Here:

- A = matrix

$$x = \text{eigenvector}$$

$$\lambda = \text{eigenvalue}$$

Simple example:

- Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

If $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then

$$Ax = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2x$$

So, x is an eigenvector and 2 is the eigenvalue.

Use of eigenvalues and eigenvectors:

- data analysis
- image compression
- machine learning
- facial recognition
- system stability analysis

6. Factorization / Decomposition (SVD, PCA, LDA)

- Factorization or decomposition means breaking a matrix or dataset into smaller meaningful parts. It helps us understand data better and reduce complexity.

6.1 Singular Value Decomposition (SVD)

- SVD breaks a matrix into three matrices.
- $A = U \Sigma V^T$

Meaning:

- U = left singular vectors
- Σ = singular values
- V^T = right singular vectors

Simple example:

- If we have a data matrix, SVD helps separate the most important information from less important information. It is useful in image compression and recommendation systems.

6.2 Principal Component Analysis (PCA)

- PCA is used to reduce the number of features in data while keeping the most important information.
- $Z = XW$

Meaning:

- X = original data
- W = principal components
- Z = reduced data

Simple example:

- Suppose a student dataset has height and weight. PCA can combine these into one main feature that explains most of the variation.

6.3 Linear Discriminant Analysis (LDA)

- LDA is used to separate data into classes. It tries to find the best line or direction that makes classes easy to distinguish.
- $W = \text{argmax} (\text{between-class variance} / \text{within-class variance})$

Simple example:

- If we have marks of two groups of students, LDA helps find a direction where the two groups look more clearly separated.

Difference between SVD, PCA, and LDA:

- SVD: matrix decomposition method
- PCA: reduces dimensions based on variance
- LDA: reduces dimensions based on class separation

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